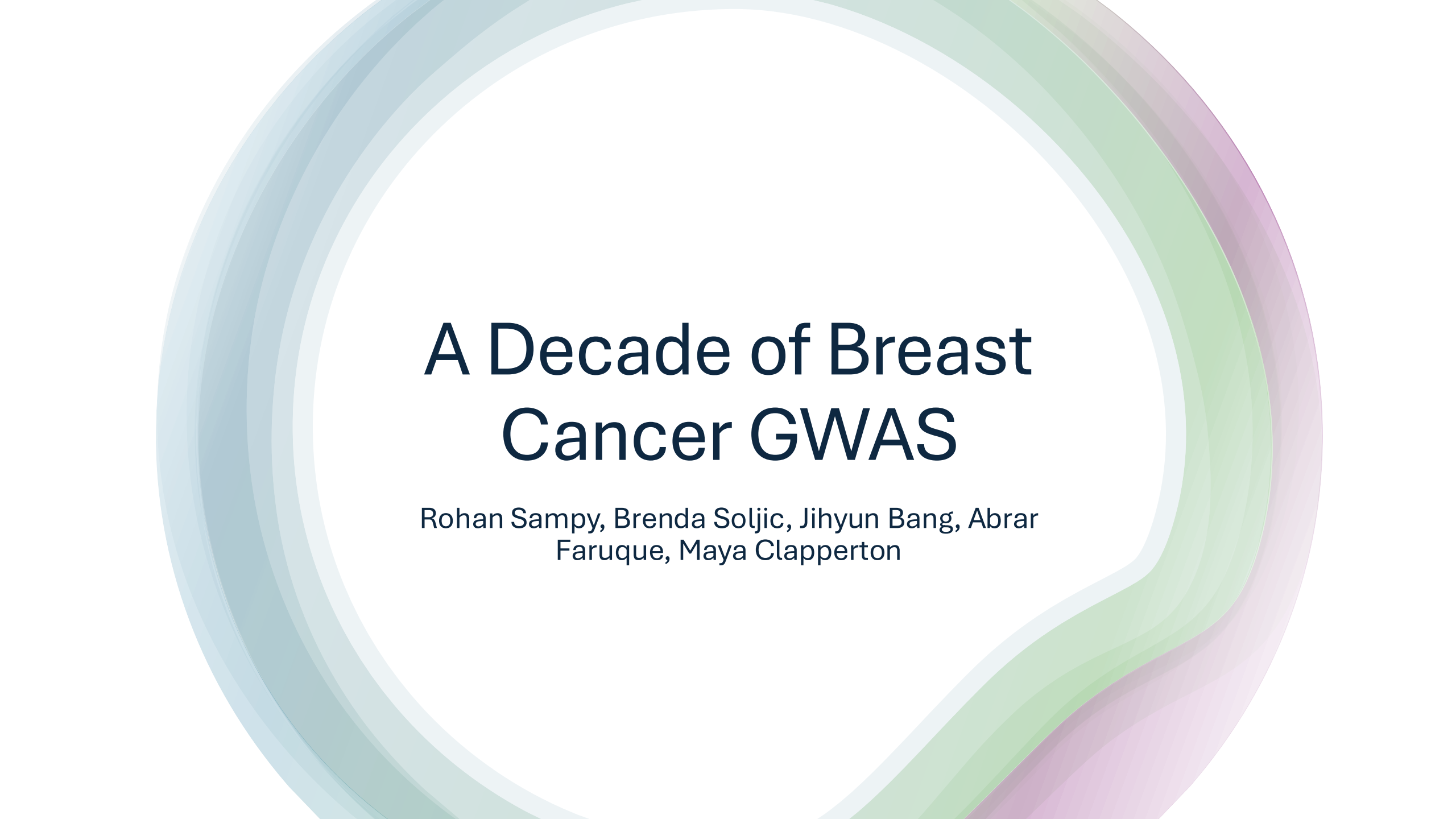




A Decade of Breast Cancer GWAS

Rohan Sampy, Brenda Soljic, Jihyun Bang, Abrar
Faruque, Maya Clapperton

Two Papers

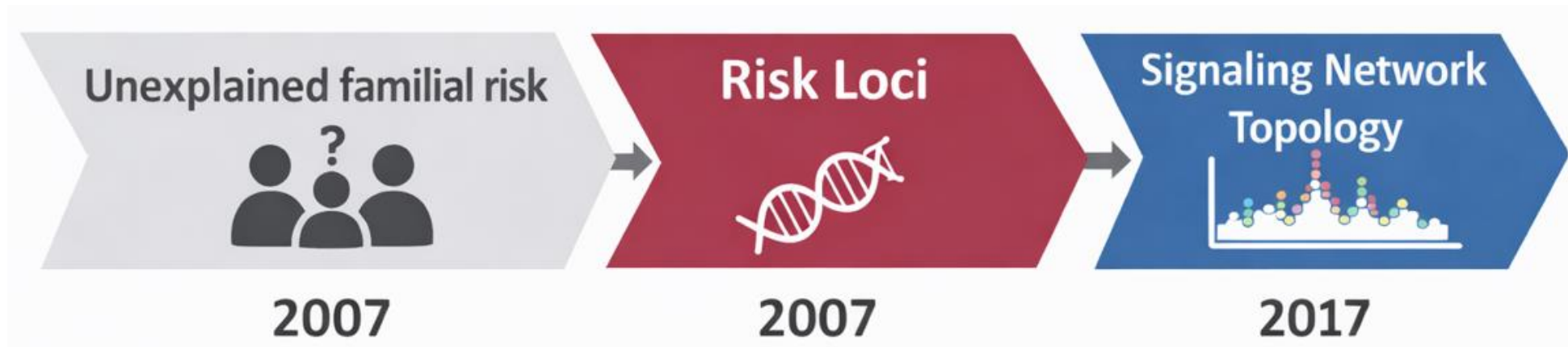


► [Nature](#). Author manuscript; available in PMC: 2009 Jul 24.

Published in final edited form as: *Nature*. 2007 Jun 28;447(7148):1087–1093. doi: [10.1038/nature05887](https://doi.org/10.1038/nature05887) [↗](#)

Genome-wide association study identifies novel breast cancer susceptibility loci

Easton et al. *Nature* 2007



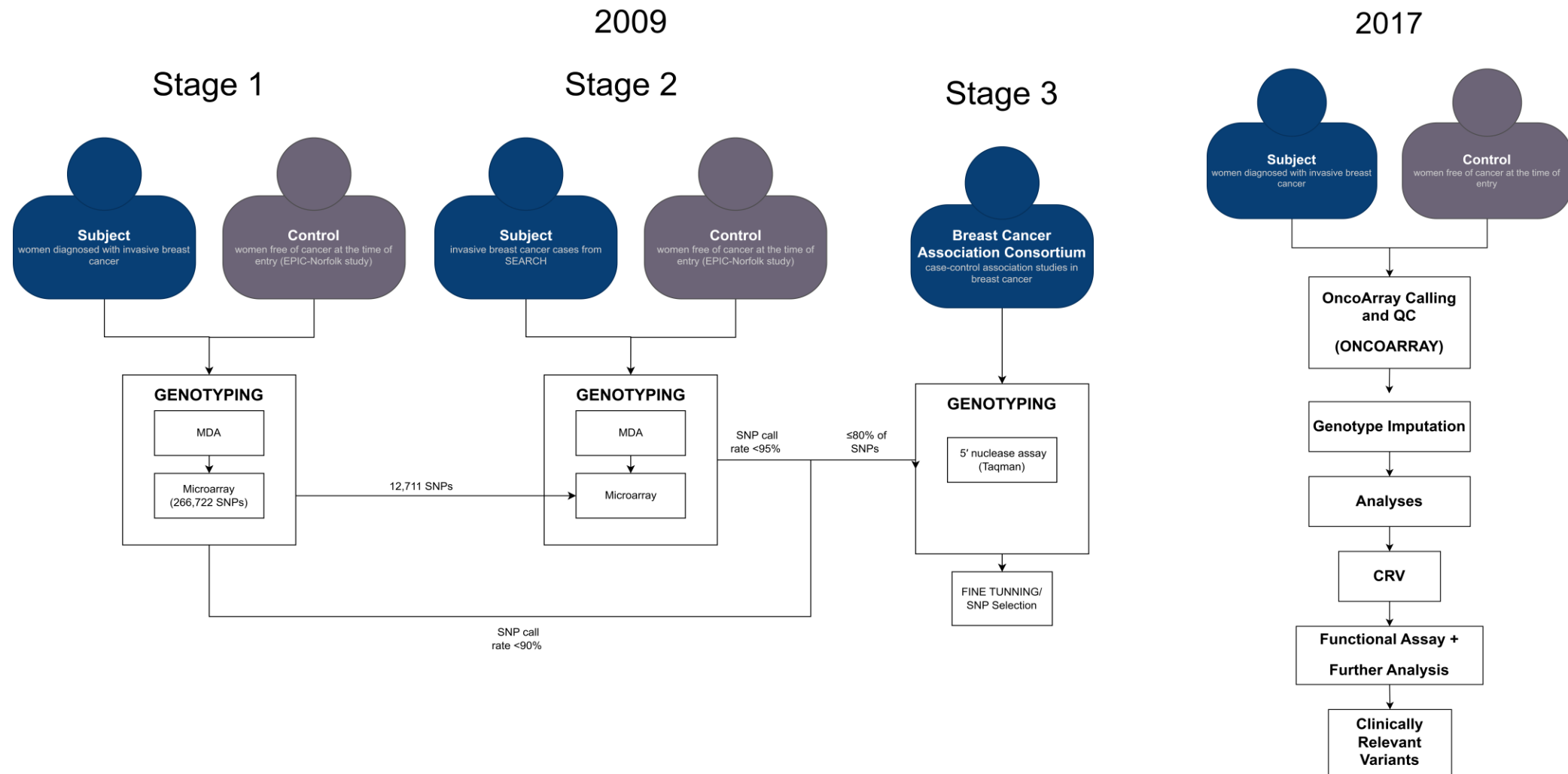
► [Nature](#). Author manuscript; available in PMC: 2018 Apr 23.

Published in final edited form as: *Nature*. 2017 Oct 23;551(7678):92–94. doi: [10.1038/nature24284](https://doi.org/10.1038/nature24284) [↗](#)

Association analysis identifies 65 new breast cancer risk loci

Michailidou et al. *Nature* 2017

Experimental Approach



Compare & Contrast

Category	2007 paper	2017 paper
Main goal	Prove GWAS can find common risk loci	Expand architecture and interpret biology
Scale	4,398/4,316 discovery, then 21,860/22,578 replication	122,977/105,974 Europeans + 14,068/13,104 East Asians
Variant coverage	227,876 SNPs	11.8M tested after imputation from ~21M variants
Output	5 new loci	65 new loci
Biology	Candidate-gene discussion, limited mechanistic inference	CRVs, regulatory annotation, INQUISIT, heritability enrichment
Population scope	Mainly European-focused, though stage 3 included diverse centers	Explicit European and East Asian analyses
Clinical framing	Common susceptibility loci exist	Better polygenic risk prediction and prevention

From Signal to Mechanism

- **2007:** strongest hit at **FGFR2**; fine-mapping suggested regulation, but causal mechanisms remained unresolved; some loci, like **8q**, were biologically opaque.
- **2017:** risk loci were interpreted through **CRVs**, **regulatory annotation**, **INQUISIT**, and targeted **functional validation**, shifting GWAS from hit-finding toward regulatory mechanism.
- The shift was not just bigger sample size. In 2007, non-coding loci were mainly association signals near plausible genes. By 2017, they were analyzed as regulatory elements connected to target genes and chromatin architecture.

What 2017 Adds That 2007 Could Not

- **ER-positive vs ER-negative** stratification
- Explicit **European and East Asian** comparison, with correlated effect sizes across populations
- Genome-wide enrichment analyses show breast cancer heritability is concentrated in **regulatory features**, including TFBS and promoter-associated marks
- Study explicitly ties findings to **polygenic risk scores** and individualized prevention

Strengths and Weaknesses

2007

Strengths

- New model for breast cancer susceptibility – polygenic inheritance
- GWAS proof of concept

Weaknesses

- Limited genomic coverage and power – only 77% of known SNPs

2017

Strengths

- Large scale and power – able to identify 65 new risk loci
- Improved functional predictions

Weaknesses

- Remaining unexplained risk – identified loci only explain 18% of familial risk

Conclusion

The 2007 paper established that GWAS could discover common breast cancer susceptibility loci.

The 2017 paper showed how the field has matured: much larger sample sizes, better variant coverage, cross-population analysis, and a major shift from locus discovery toward interpretation and risk prediction.